

WES-237C Project 2 - CORDIC Design Report

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Measurement basis. Results are from Vitis HLS 2024.2 synthesis reports and C-testbench runs. Throughput is estimated from the synthesized clock period and top-level initiation interval (II):

$$R\ (\text{MSamples/s}) = \frac{1000}{T_{\text{est}}\ (\text{ns})\, I_{\text{top}}\ (\text{cycles})}. \tag{1}$$

These are HLS estimates, not hardware measurements. Radius RMSE is in radius units and angle RMSE is in radians.

Q1 Floating-Point CORDIC: Rotation Count

Q1.(a) Resource, Throughput, Latency, and Error

We implemented an iterative floating-point CORDIC with 10, 12, 14, 16, 18, and 20 rotations. The 12-rotation variant was corrected to execute exactly 12 iterations, and its C-test and synthesis results were updated. To improve functional coverage, the testbench was expanded from four to **22 deterministic inputs**, including axis, quadrant, small-magnitude, and boundary cases.

Rot.	Lat. (cyc.)	Top II (cyc.)	Clock (ns)	Rate (MSa/s)	BRAM	DSP	FF	LUT	RMSE r	RMSE θ (rad)
10	110	111	7.256	1.242	0	10	1,524	2,412	1.01e-06	1.04e-03
12	130	131	7.256	1.052	0	10	1,524	2,414	5.97e-08	3.09e-04
14	150	151	7.256	0.913	0	10	1,524	2,416	7.34e-08	6.28e-05
16	170	171	7.256	0.806	0	10	1,524	2,418	7.34e-08	1.79e-05
18	190	191	7.256	0.722	0	10	1,525	2,420	7.34e-08	5.77e-06
20	210	211	7.256	0.653	0	10	1,525	2,422	7.34e-08	1.31e-06

Table 1: Q1(a): floating-point CORDIC versus rotation count.

Increasing rotations from 10 to 20 raised latency from **110 to 210 cycles** and reduced estimated throughput from **1.242 to 0.653 MSamples/s** (47.4%). Over the same range, phase RMSE improved from 1.04×10^{-3} to 1.31×10^{-6} rad. DSP usage remained at 10, while LUT usage rose only from 2,412 to 2,422. The 10-rotation phase result narrowly exceeds the testbench's 0.001 RMSE threshold.

Q1.(b) Measured Trends versus Rotation Count

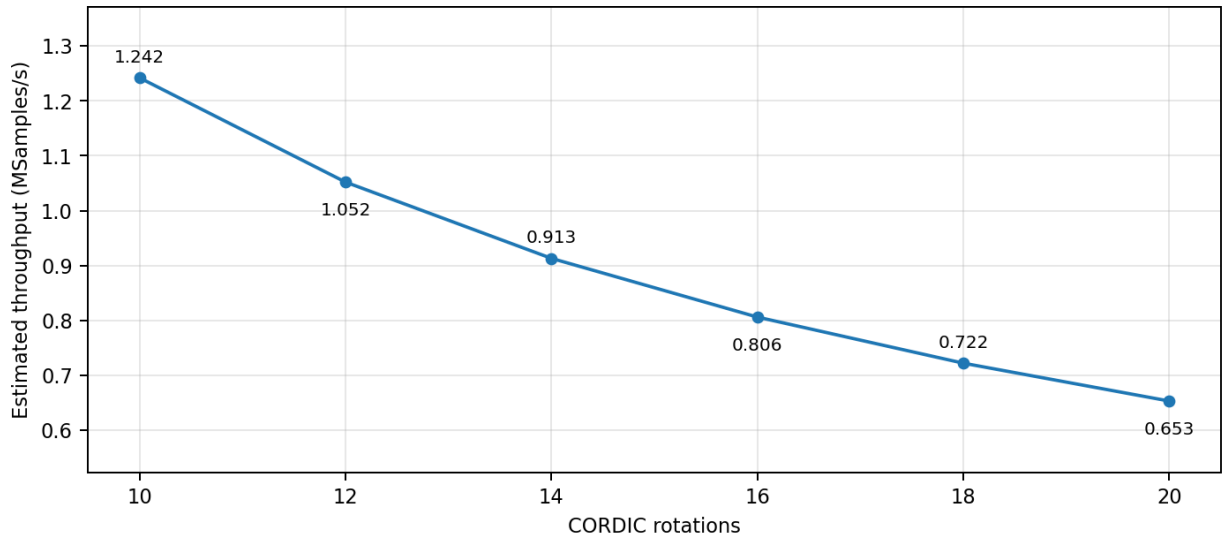


Figure 1: Q1(b): estimated throughput versus rotation count.

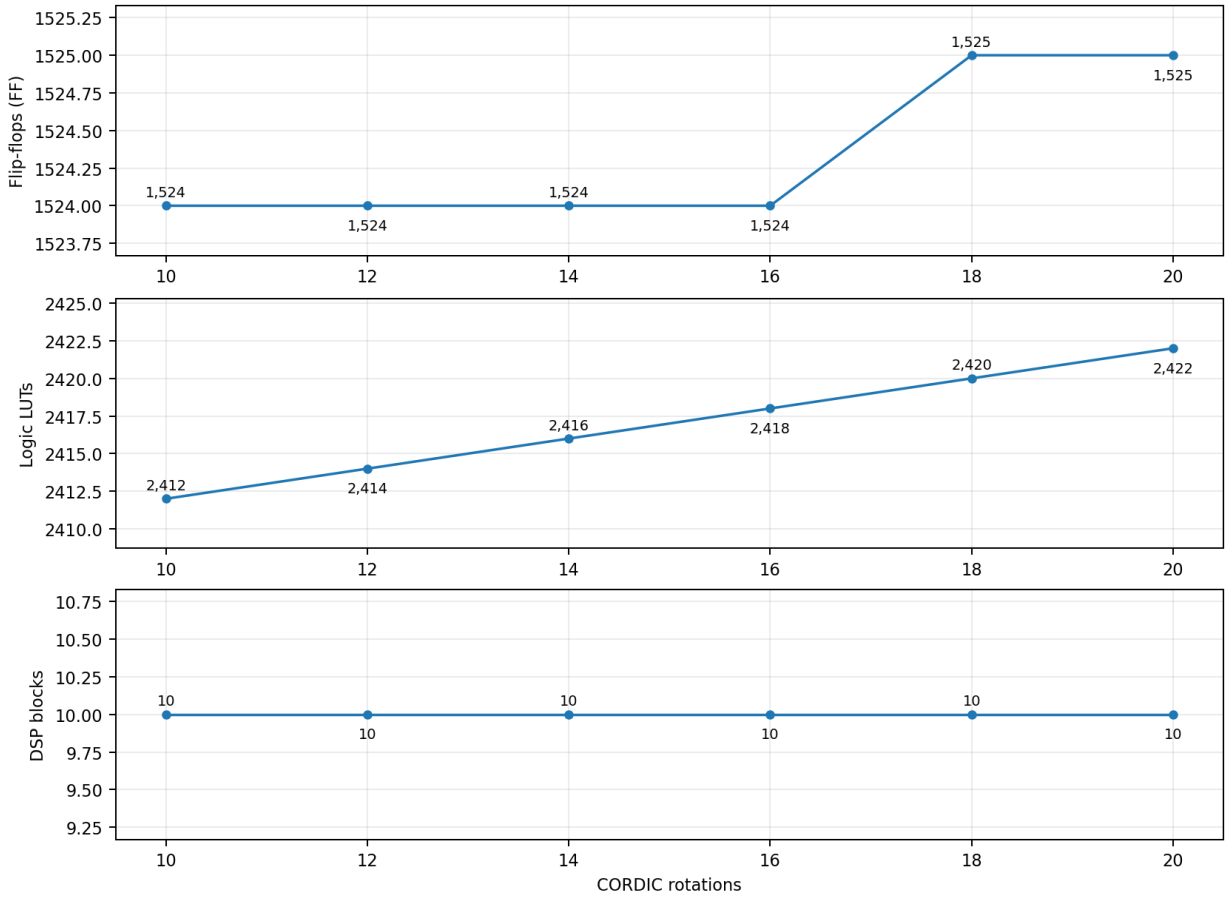


Figure 2: Q1(b): FF, logic LUT, and DSP usage versus rotation count.

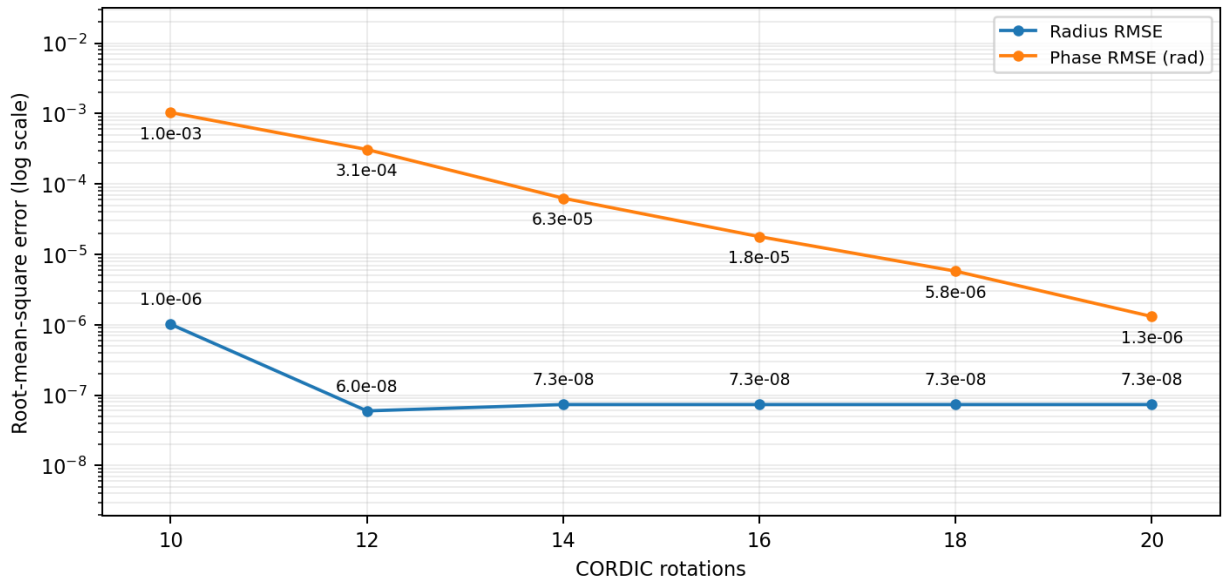


Figure 3: Q1(b): radius and phase RMSE versus rotation count (logarithmic scale).

Q1.(c) Accuracy Saturation

Radius accuracy effectively levels off at approximately **12 rotations**, with RMSE near 10^{-7}. Phase accuracy continues improving through 20 rotations, so its saturation point was **not reached** within this sweep.

Q2 Arbitrary-Precision Data Types

Q2.(a) Required Integer Widths

Signed fixed-point types include the sign bit in the integer width. For normalized $x, y \in [-1, 1]$, the required widths are 2 integer bits for each coordinate and radius, and 3 bits for angle and intermediate rotated coordinates. We therefore use the largest requirement, **3 integer bits**, in the common-width experiments.

Quantity	Integer bits	Range / rationale
Cartesian inputs x, y	2	$[-1, 1]$
Output radius r	2	$[0, \sqrt{2}]$
Output phase θ	3	$[-\pi, \pi]$
Internal rotated coordinates	3	Approximate magnitude up to 2.33

Table 2: Q2(a): signed integer-width requirements for the tested domain.

For Q2 and Q3, the function input/output arguments remain `float`; conversions to and from `ap_fixed` occur inside the function. The reported synthesis results include those conversions.

Q2.(b) Common Fixed-Point Bitwidth Sweep

With 16 rotations and 3 integer bits fixed, the internal arithmetic and table widths were swept over 8, 12, 16, 20, 24, and 32 total bits.

Bits	Lat. (cyc.)	Top II (cyc.)	Clock (ns)	Rate (MSa/s)	BRAM	DSP	FF	LUT	RMSE r	RMSE θ (rad)
8	42	43	7.182	3.238	0	0	836	2,419	3.54e-02	6.34e-01
12	44	45	7.287	3.050	0	3	982	2,413	9.93e-04	8.88e-03
16	44	45	7.181	3.095	0	3	1,261	3,307	5.87e-05	6.11e-04
20	45	46	7.177	3.029	0	3	1,332	3,427	5.51e-06	1.32e-04
24	45	46	7.282	2.985	0	6	1,684	4,114	3.65e-07	1.81e-05
32	79	80	7.164	1.745	0	12	2,426	4,712	2.22e-08	1.79e-05

Table 3: Q2(b): corrected float-interface / internally fixed-point bitwidth sweep.

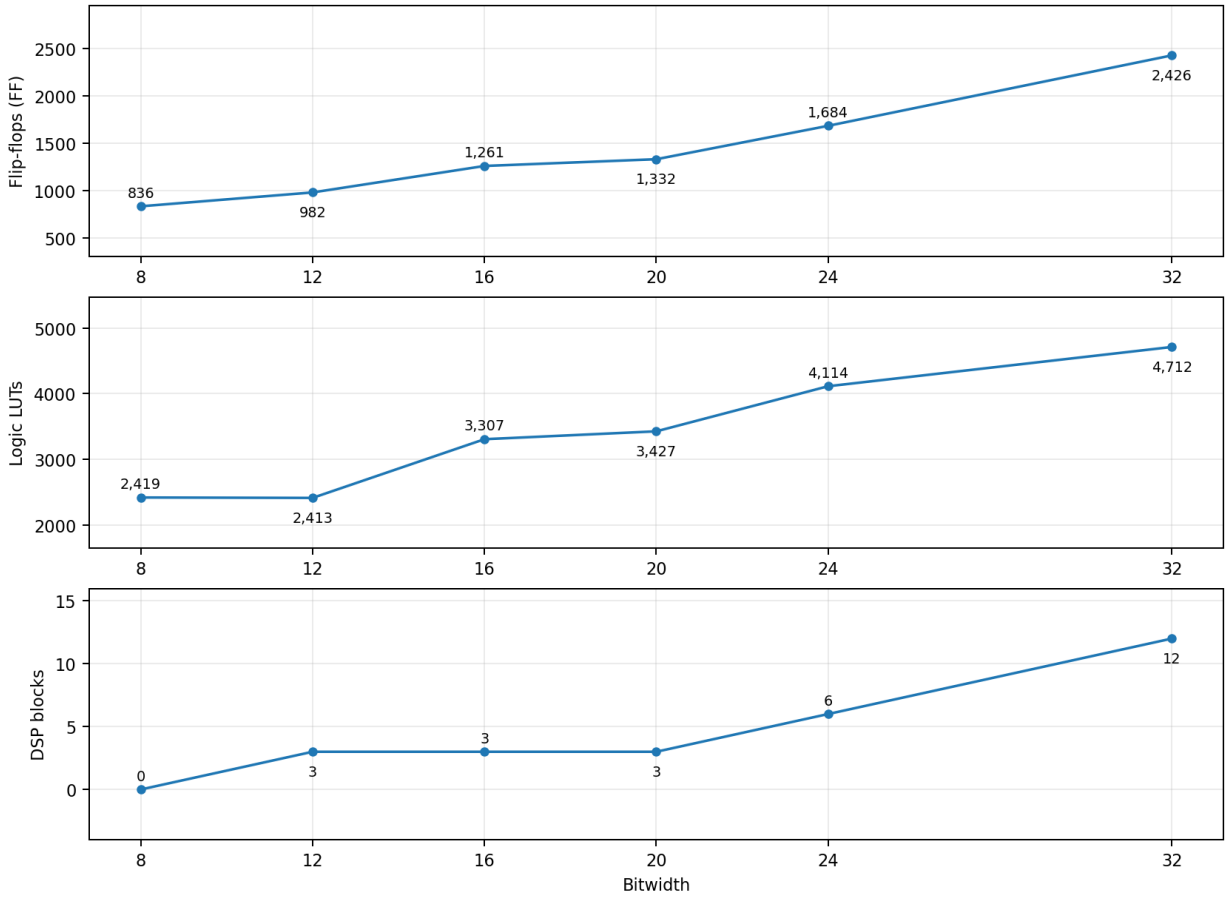


Figure 4: Q2(b): resource utilization versus internal fixed-point width.

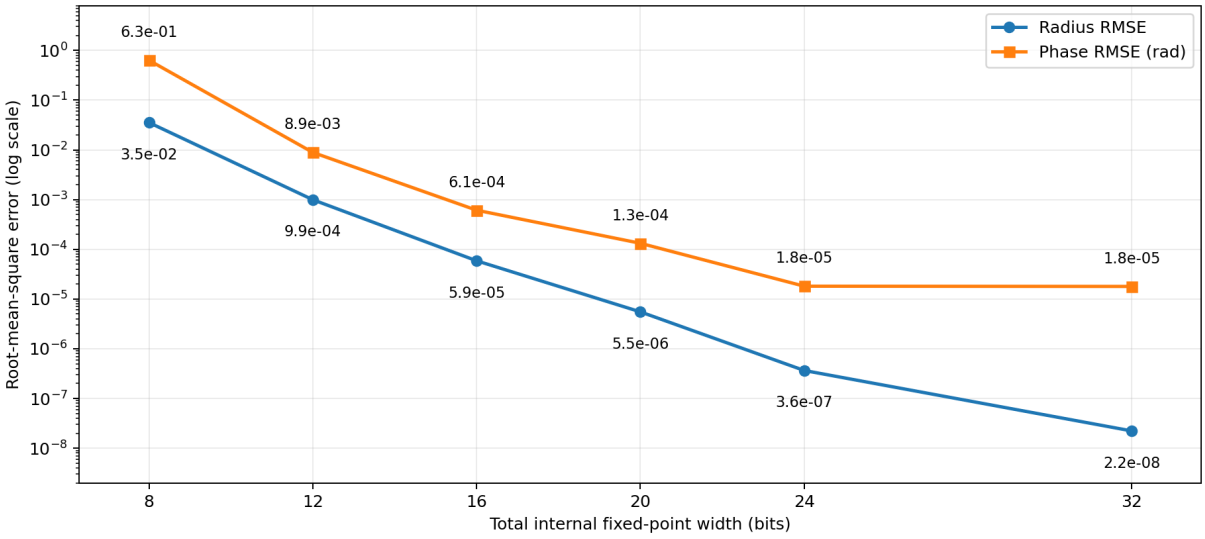


Figure 5: Q2(b): radius and phase RMSE versus internal fixed-point width.

At 16 bits, throughput reached **3.095 MSamples/s**, compared with **0.806 MSamples/s** for the 16-rotation floating-point baseline (3.84× higher), while DSP usage decreased from 10 to 3. Greater width improves accuracy but generally consumes more resources: at 32 bits the design uses 12 DSPs and achieves 1.745 MSamples/s. Radius RMSE continues decreasing through 32 bits, whereas phase RMSE stabilizes near 1.8×10^{-5} rad at 24–32 bits. The 8- and 12-bit variants exceed the 0.001 phase-RMSE threshold.

Q2.(c) Rotation Table Bitwidth Sweep

The internal coordinates were held at `ap_fixed<16,3>` while only the gain and angle table widths changed. The required 4–32 bit configurations were tested, with an additional 24-bit measurement.

Bits	Lat. (cyc.)	Top II (cyc.)	Clock (ns)	Rate (MSa/s)	BRAM	DSP	FF	LUT	RMSE r	RMSE θ (rad)
4	44	45	7.181	3.095	0	1	1,215	3,309	5.01e-02	5.44e-01
8	44	45	7.181	3.095	0	3	1,230	3,303	1.50e-04	1.55e-02
12	44	45	7.181	3.095	0	3	1,245	3,305	5.87e-05	2.15e-03
16	44	45	7.181	3.095	0	3	1,261	3,307	5.87e-05	6.11e-04
20	44	45	7.181	3.095	0	3	1,287	3,363	5.87e-05	6.08e-04
24	44	45	7.181	3.095	0	3	1,303	3,409	5.87e-05	6.08e-04
32	60	61	7.181	2.283	0	5	1,572	3,490	5.87e-05	6.08e-04

Table 4: Q2(c): rotation-table bitwidth sweep with 16-bit internal coordinates.

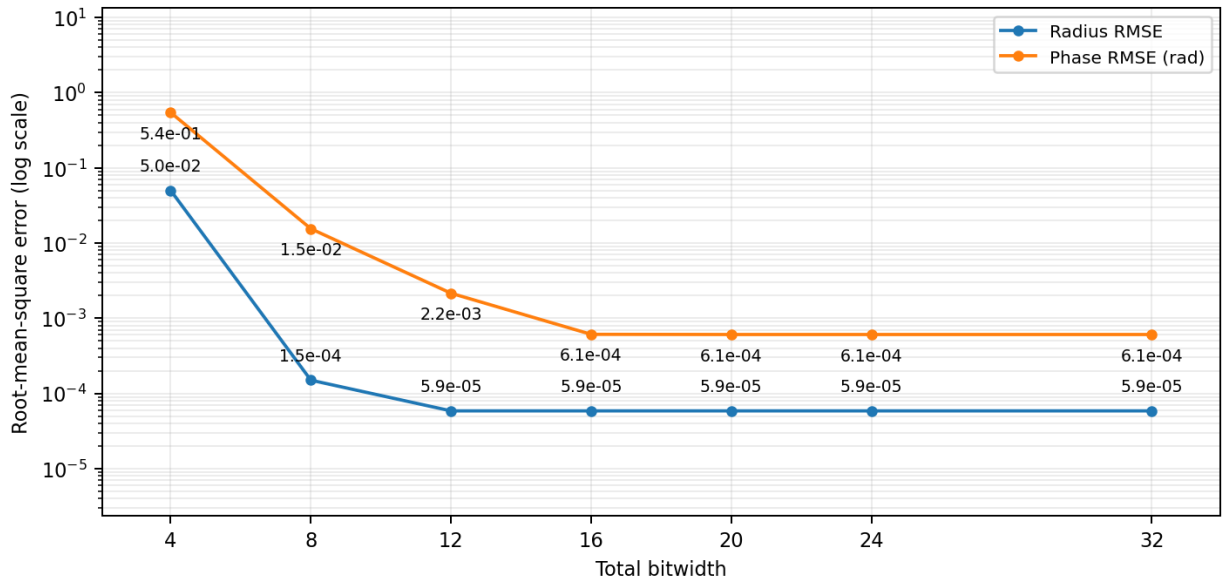


Figure 6: Q2(c): radius and phase RMSE versus rotation-table width.

Radius RMSE levels off near 5.87×10^{-5} by **12 table bits**; phase RMSE levels off near 6.1×10^{-4} rad by **16–20 bits**. Increasing table precision to 32 bits raises DSP use from 3 to 5 and top-level II from 45 to 61 cycles, without a meaningful accuracy improvement. The 4-, 8-, and 12-bit tables exceed the phase-RMSE threshold.

Q3 Shift-and-Add CORDIC

Q3.(a) Multiplier-Based versus Shift-Based Operations

We replaced per-rotation multiplications with fixed-point shifts and additions, retaining the permitted final gain correction. Each configuration uses the same internal width and floating-point interface as its Q2(b) baseline.

Bits	Lat. (cyc.)	Top II (cyc.)	Clock (ns)	Rate (MSa/s)	BRAM	DSP	FF	LUT	RMSE r	RMSE θ (rad)
8	42	43	7.182	3.238	0	0	820	2,334	1.37e-01	6.34e-01
12	43	44	7.287	3.119	0	1	958	2,430	8.36e-03	8.74e-03
16	42	43	7.181	3.239	0	1	1,228	3,315	4.43e-04	6.35e-04
20	43	44	7.177	3.167	0	1	1,291	3,446	1.77e-05	2.16e-04
24	43	44	7.282	3.121	0	2	1,635	4,104	1.02e-06	2.59e-05
32	45	46	7.164	3.034	0	4	1,901	4,706	1.94e-08	1.79e-05

Table 5: Q3(a): corrected float-interface shift-and-add implementations.

At 16 bits, shift-and-add reduced DSP usage from **3 to 1** and increased estimated throughput from **3.095 to 3.239 MSamples/s**. FFs decreased from 1,261 to 1,228, while LUT usage stayed nearly constant (3,307 versus 3,315). Radius RMSE increased from 5.87×10^{-5} to 4.43×10^{-4}, illustrating an accuracy tradeoff. At 32 bits, DSPs fell from 12 to 4 and throughput increased from 1.745 to 3.034 MSamples/s.

Q3.(b) Comparison of LUTs, DSPs, and FFs

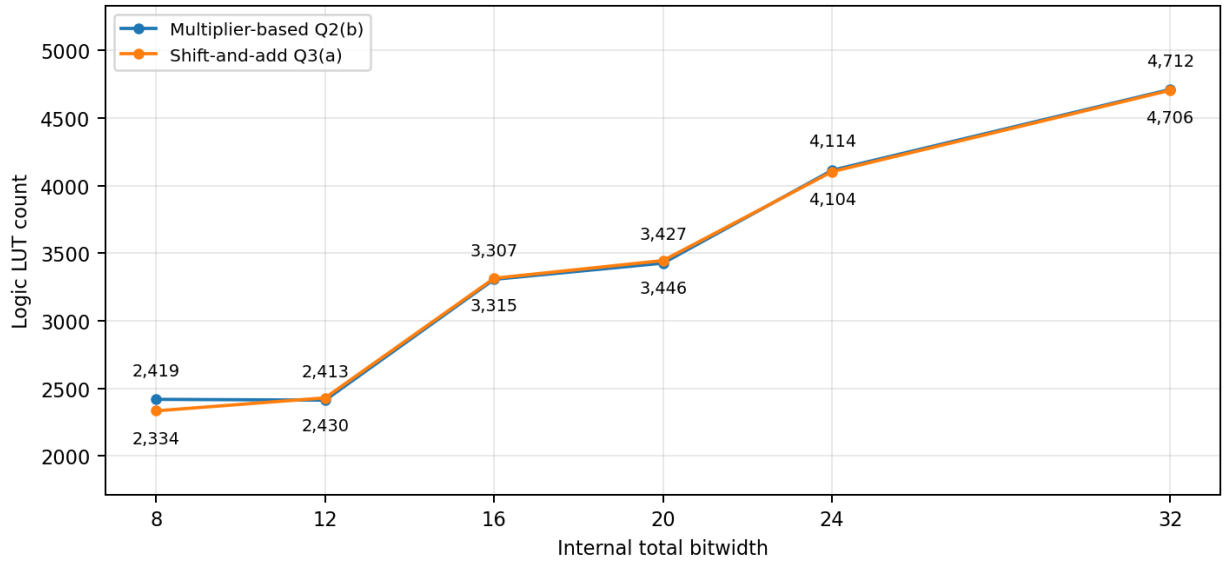


Figure 7: Q3(b): LUT utilization for multiplication and shift-and-add.

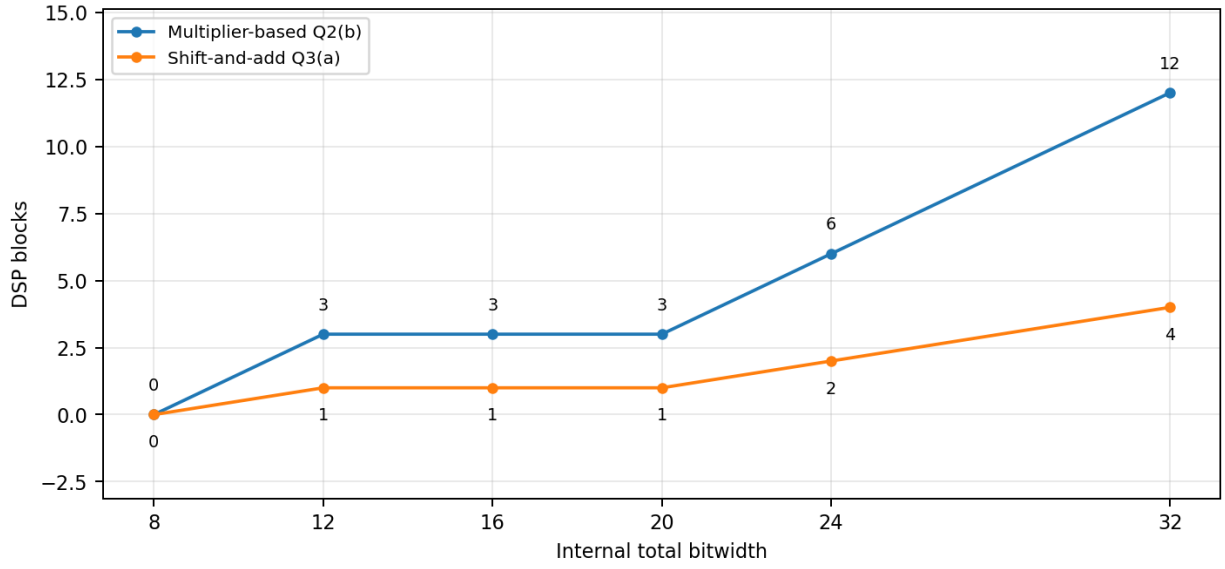


Figure 8: Q3(b): DSP utilization for multiplication and shift-and-add.

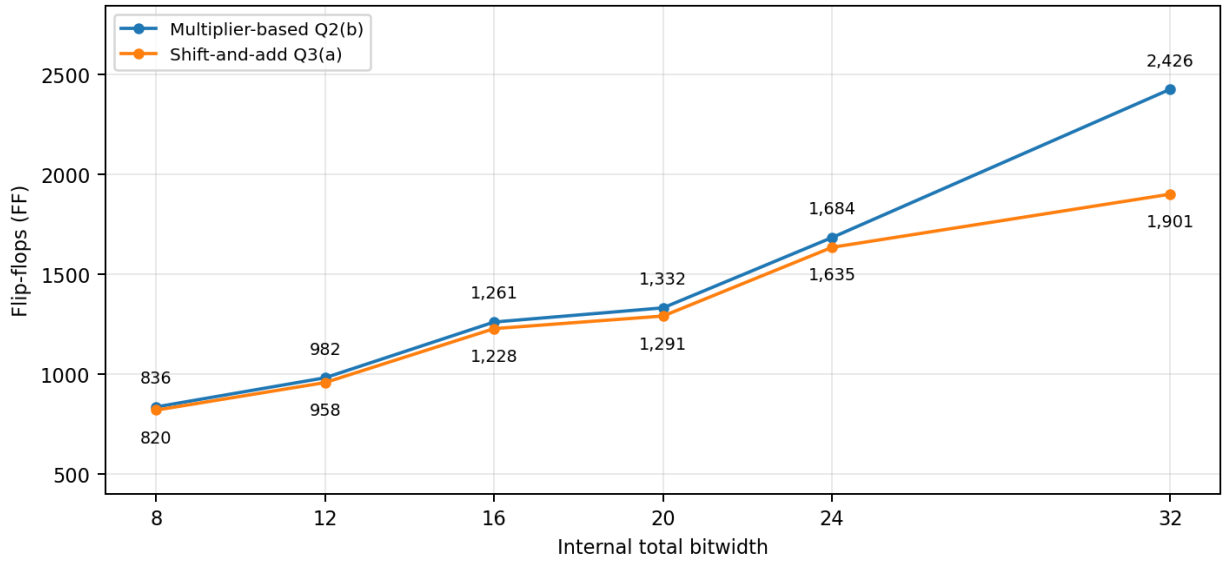


Figure 9: Q3(b): FF utilization for multiplication and shift-and-add.

Shift-and-add consistently reduces DSP consumption for the 12–32 bit designs. FF savings are generally modest, and LUT counts remain similar; eliminating multipliers therefore provides the clearest benefit in DSP utilization.

Q4 Lookup-Table Cartesian-to-Polar Architecture

Q4.(a) LUT Storage as a Function of Word Width

The LUT architecture addresses stored radius and angle values using the combined bit patterns of the two inputs. For input widths B_x, B_y and output widths B_r, B_θ, the raw storage requirement is

$$N_{\text{bits}} = 2^{B_x+B_y}(B_r + B_\theta). \tag{2}$$

Thus, storage grows **exponentially with input width** and **linearly with output width**. For equal widths W on all four values, capacity is $2^{2W}(2W)$ bits, split between radius and angle tables.

Width W	Addresses 2^{2W}	Bits per pair	Raw table bits
4	256	8	2,048
5	1,024	10	10,240
6	4,096	12	49,152
7	16,384	14	229,376
8	65,536	16	1,048,576

Table 6: Q4(a): theoretical combined radius and phase ROM capacity.

Q4.(b) LUT Bitwidth Sweep

The separate direct-address ROM implementation was tested at 4, 5, 6, 7, and 8 total bits with 3 integer bits per value. Its testbench compares outputs with floating-point reference values on a **201×201 Cartesian grid**.

Bits	Lat. (cyc.)	Top II (cyc.)	Clock (ns)	Rate (MSa/s)	BRAM	DSP	FF	LUT	RMSE r	RMSE θ (rad)
4	1	2	3.254	153.657	2	0	2	14	1.83e-01	1.39e+00
5	1	2	3.254	153.657	2	0	2	14	9.23e-02	1.04e+00
6	1	2	3.254	153.657	4	0	2	14	4.87e-02	7.42e-01
7	1	2	3.254	153.657	14	0	2	14	2.46e-02	5.32e-01
8	1	2	3.254	153.657	64	0	2	14	1.24e-02	3.12e-01

Table 7: Q4(b): direct-address LUT implementation, 4–8 total bits.

All five designs have **1-cycle latency**, a **2-cycle top-level II**, and estimated throughput of **153.657 MSamples/s**. DSP and logic LUT counts remain at 0 and 14, respectively. BRAM usage rises from **2 blocks at 4 bits** to **64 blocks at 8 bits**, reflecting the rapid growth in stored entries.

Q4.(c) LUT Resource Consumption

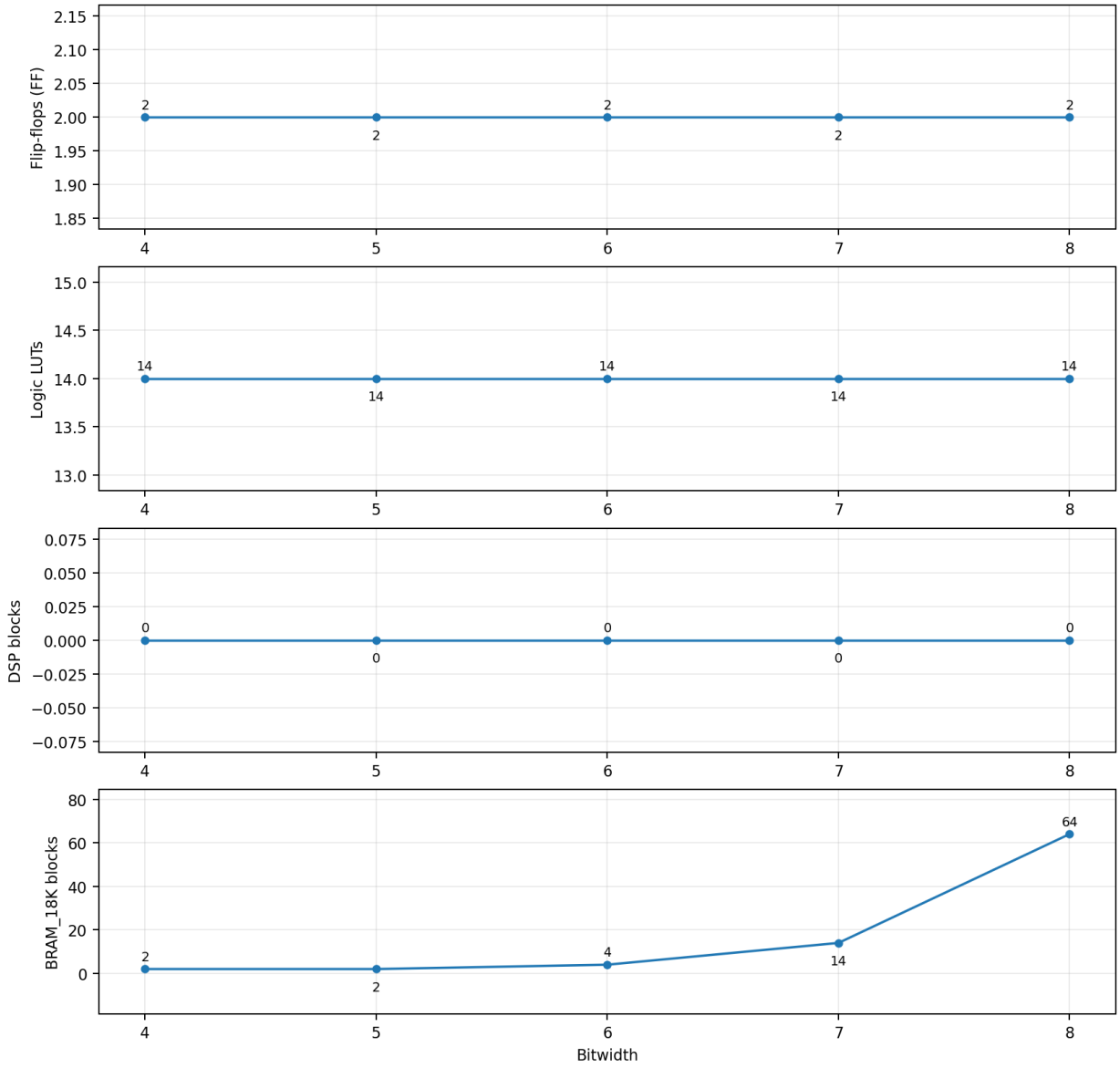


Figure 10: Q4(c): LUT, FF, DSP, and BRAM usage versus word width.

The dominant resource cost is **BRAM**, not logic LUTs: the synthesis maps the stored radius and angle tables to embedded memory. FF, DSP, and logic LUT usage changes little over the sweep.

Q4.(d) LUT Accuracy versus Word Width

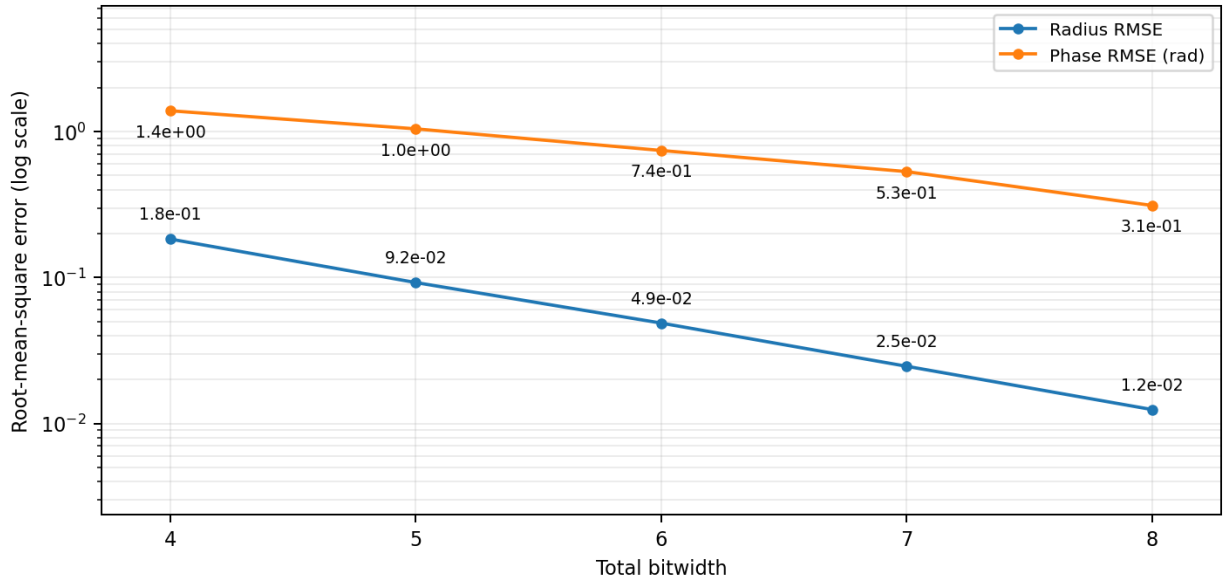


Figure 11: Q4(d): radius and phase RMSE versus LUT word width (logarithmic scale).

Radius RMSE decreases from **0.183 to 0.0124** and phase RMSE from **1.39 to 0.312 rad** as width increases from 4 to 8 bits. Higher precision improves both outputs, but input quantization limits accuracy at these widths.

Q4.(e) CORDIC versus LUT Architectural Tradeoffs

The LUT delivers much lower latency and higher estimated throughput: at 8 bits it achieves **153.657 MSamples/s** with zero DSPs, but consumes **64 BRAM blocks**. The 16-bit shift-and-add CORDIC achieves **3.239 MSamples/s** using 1 DSP and no BRAM, with substantially finer internal precision. **CORDIC favors memory efficiency and adjustable numerical precision; lookup favors speed at the cost of exponential memory growth.** The two RMSE testbenches use different input sets, so their absolute RMSE results are not a controlled head-to-head accuracy comparison.

GitHub Repositories

- **Joshua Hoang:** https://github.com/HoangJ25/WES237C_Read_the_Docs/tree/master/project_files/project2
- **Ryan Shimizu:** https://github.com/ryan-shimizu/Read_the_docs/tree/rshimizu-project2/project_files/project2